

Methods

1. Dataset Preparation

The dataset comprised 55,490 RGB training image patches (28×28 pixels) and 4,367 independent test images extracted from haematoxylin- and eosin-stained histological whole-slide images. Five tissue classes were included: adipose, lymphocytes, normal colon mucosa, cancer-associated stroma, and colorectal adenocarcinoma epithelium. The training data were partitioned into 67 % training and 33 % validation subsets using stratified sampling to preserve class proportions. Pixel intensities were normalised using the mean and standard deviation calculated from the training set to prevent data leakage and ensure consistent feature scaling across validation and test data. For the baseline model, each image was flattened into a 2,352-dimensional vector, whereas the convolutional neural network (CNN) received tensors in the format (N, 3, 28, 28).

2. Baseline Logistic Regression Model

A multinomial logistic regression classifier was implemented as a baseline to establish a lower-bound performance estimate. Logistic regression models linear decision boundaries and does not capture spatial relationships between pixels, making it suitable for evaluating the benefit of convolutional feature extraction in image classification.

3. Convolutional Neural Network (CNN) Architecture and Training

The CNN architecture comprised two convolutional blocks, each including convolution, ReLU activation, and max-pooling layers, followed by a fully connected classifier (Figure 1). Convolutional layers learn hierarchical spatial features such as texture and cellular morphology, while pooling reduces dimensionality and provides translation invariance, which are desirable properties for histological image analysis.

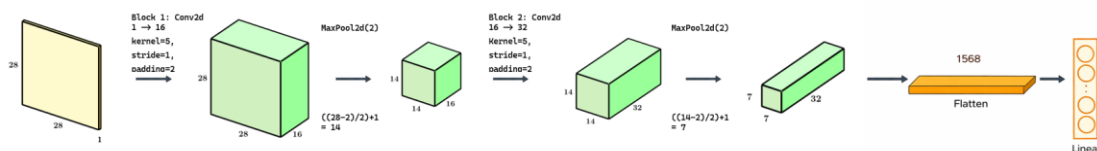


Figure 1. Architecture of the convolutional neural network used in this study.

Models were trained using the Adam optimiser with learning rate = 0.001 and CrossEntropyLoss for up to 20 epochs ^[1-2] with early stopping based on validation accuracy. A grid search evaluated combinations of batch sizes (32, 64), learning rates (0.001, 0.0001), epochs (10, 15), dropout (0.3, 0.5), and network depth (2, 3 convolutional blocks). The optimal configuration was selected using validation accuracy and the train-validation gap to assess overfitting. All experiments were

implemented in PyTorch with deterministic seeds where possible.

4. Evaluation

Model performance was evaluated on an independent test dataset using overall accuracy and confusion matrices to provide an unbiased estimate of generalisation behaviour across tissue classes.

Result

1. Data exploration

Figure 2 shows that the training dataset was approximately balanced across the five tissue classes, whereas the test set displayed moderate imbalance, with adipose and adenocarcinoma epithelium comprising 58.8 % of samples. Representative image patches from each category are illustrated in Figure 3. Normalisation produced similar mean and variance in the training (mean = 0.0000, SD = 1.0000) and validation sets (mean = -0.0016, SD = 1.0011), indicating comparable distributions. In contrast, the test set showed a shift (mean = 0.1121, SD = 0.9115), suggesting domain variation between slide regions that may influence generalisation performance.

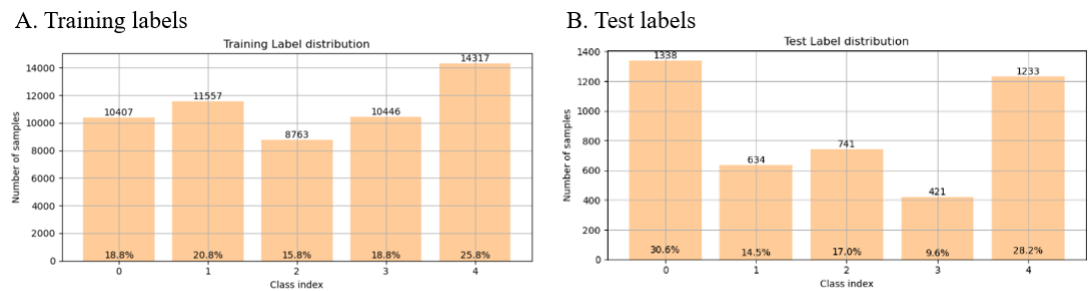


Figure 2. Distribution of class labels in the training and test datasets.
0: adipose, 1: lymphocytes, 2: normal colon mucosa, 3: cancer-associated stroma, 4: colorectal adenocarcinoma epithelium

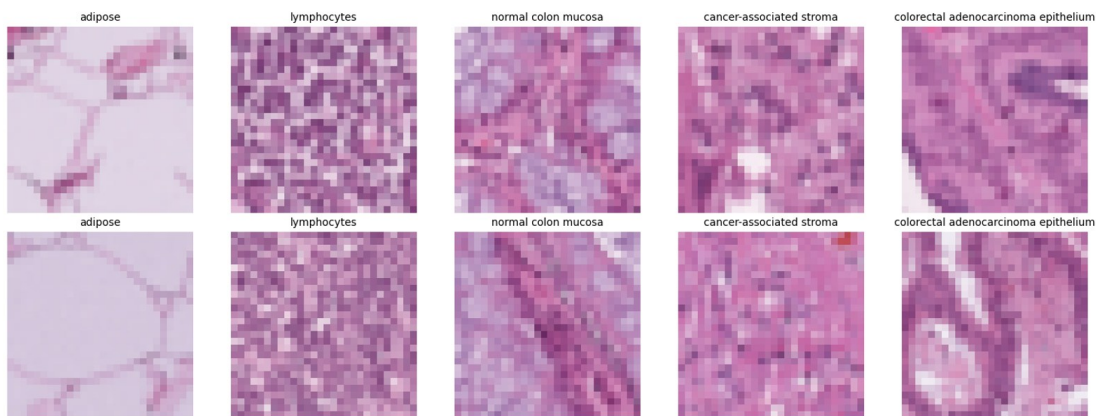


Figure 3. Representative sample images from each class.

2. Baseline Logistic Regression Model

Table 1. Classification Report for the logistic regression model on the test dataset.

	precision	recall	f1-score
adipose	0.10	0.03	0.05
lymphocytes	0.19	0.20	0.19
normal colon mucosa	0.26	0.12	0.16
cancer-associated stroma	0.18	0.29	0.22
colorectal adenocarcinoma epithelium	0.27	0.50	0.35
accuracy	-	-	0.23
macro avg	0.20	0.23	0.20
weighted avg	0.20	0.23	0.19

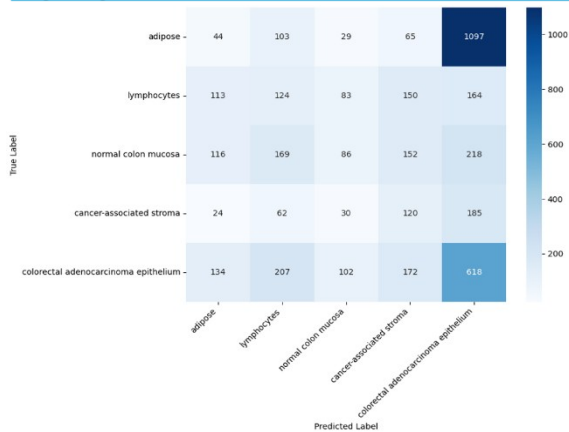


Figure 4. Confusion matrix of the baseline logistic regression model showing predicted versus true labels on the test dataset.

The multinomial logistic regression model achieved 31.26% validation accuracy and 22.72% test accuracy, barely exceeding the 20% expected from random guessing. Precision and recall were consistently low across all tissue types (Table 1), and the confusion matrix (Figure 4) revealed extensive misclassification. Adipose tissue was frequently predicted as colorectal adenocarcinoma epithelium, indicating that linear models cannot capture the complex spatial patterns characteristic of histological structures.

3. CNN Model

The CNN reached 87.12% validation accuracy (training loss = 0.3056; training accuracy = 89.01%; validation loss = 0.3463) and 87.25% test accuracy, demonstrating a substantial improvement over logistic regression. The best-performing configuration used a batch size of 32, 10 epochs, learning rate = 0.001, dropout = 0.3, and three convolutional layers. Training loss decreased steadily while validation accuracy plateaued after approximately ten epochs (Figure 5), and the small train-validation gap suggested reasonable generalisation without severe overfitting.

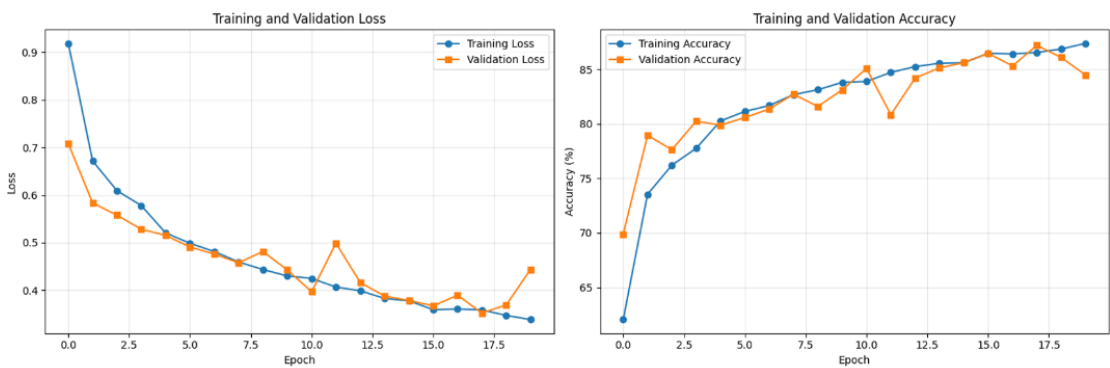


Figure 5. Training and validation trends showing loss and accuracy across epochs.

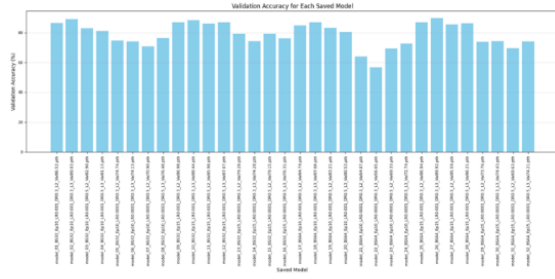


Figure 6. Validation accuracy of all CNN configurations visualised as a histogram.

every few configurations reached a new lower point, but the overall loss curve remained spiky rather than smooth. Spiky behaviour also been observed in the accuracy curves; however, the overall trend is flatted, with the small gap between train and validation data and validation getting higher accuracy.

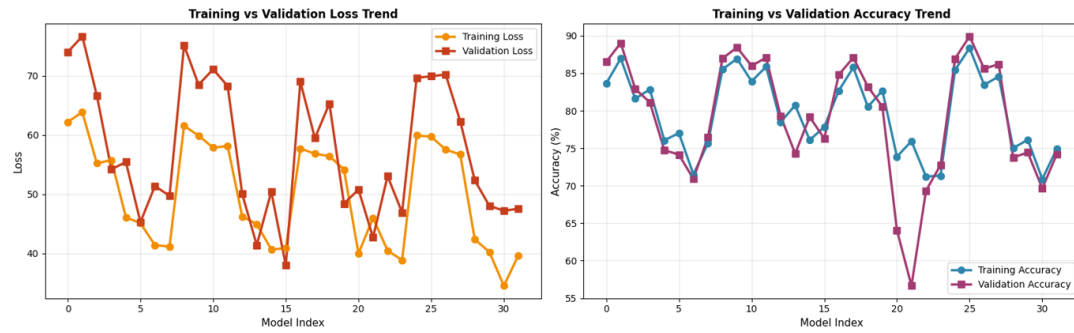


Figure 7. Training and validation trends showing loss and accuracy across all CNN configurations.

4. Evaluation

Table 2. Classification report for the best CNN model on the test dataset.

	precision	recall	f1-score
adipose	0.98	1.00	0.99
lymphocytes	0.76	0.92	0.83
normal colon mucosa	0.86	0.81	0.83
cancer-associated stroma	0.81	0.69	0.75
colorectal adenocarcinoma epithelium	0.84	0.82	0.83
accuracy	-	-	0.87
macro avg	0.85	0.85	0.85
weighted avg	0.87	0.87	0.87

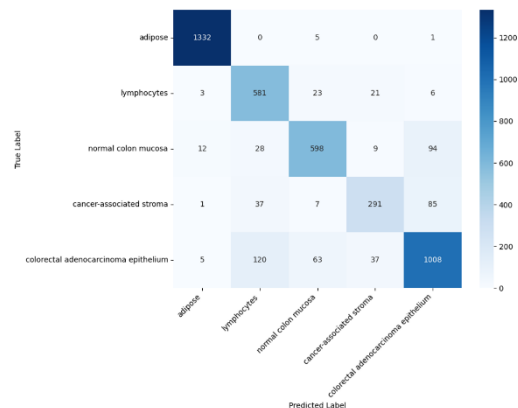


Figure 8. Confusion matrix of the best CNN model showing predicted versus true labels on the test dataset.

of 0.69 for cancer-associated stroma indicates that 31% of true mucosa samples were misidentified. Consistent with these values, misclassification was most frequent for normal colon mucosa (6.4%) and stroma (10.3%), both of which were often predicted as colorectal adenocarcinoma epithelium.

Across the full grid search (Figures 6-7), most configurations achieved validation accuracies between 70% and 90%, with two outliers showing sharp drops to around 60%. The loss values showed a more recognisable downward pattern, where

Compared with the logistic regression baseline, the CNN achieved far superior performance, reaching 87.25% test accuracy. As shown in Figure 8, adipose and lymphocyte tissues were classified reliably, whereas normal mucosa and stroma exhibited moderate confusion with colorectal adenocarcinoma epithelium. Table 2 summarises the final test metrics: precision ranged from 0.76 to 0.98 and recall from 0.69 to 1.00, yielding F1-scores between 0.75 and 0.99. The lower recall values correspond to higher misclassification rates; for example, a recall

Discussion

The CNN substantially outperformed logistic regression, achieving 87.25% compared with 22.72% test accuracy. This improvement is expected because convolutional layers learn spatial patterns that are critical for histological image classification, whereas logistic regression treats pixels independently and cannot model structural relationships. Training and validation accuracies of 89% and 87% respectively suggest that the model did not suffer from severe overfitting.

In Figure 7, loss and accuracy show differing variability across hyperparameter configurations. This behaviour is consistent with the fact that cross-entropy loss provides a differentiable measure of prediction quality, while accuracy changes only when predictions cross class decision boundaries ^[3-4]. Moreover, fluctuation been observed is more likely due to optimisation factors such as learning-rate selection, small batch size, or insufficient training epoch, which is limited by the computational cost of grid search ^[5].

Although a test accuracy of 87% is strong for an experimental model, clinical diagnostic applications typically require extremely high sensitivity and reliability. Recall is particularly important because misclassification may lead to inappropriate treatment decisions. Therefore, the CNN should be considered a decision-support tool rather than a standalone diagnostic system. Future work could include transfer learning using pretrained CNNs, and more extensive hyperparameter optimisation with greater computational resources.

Reference (67 words)

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